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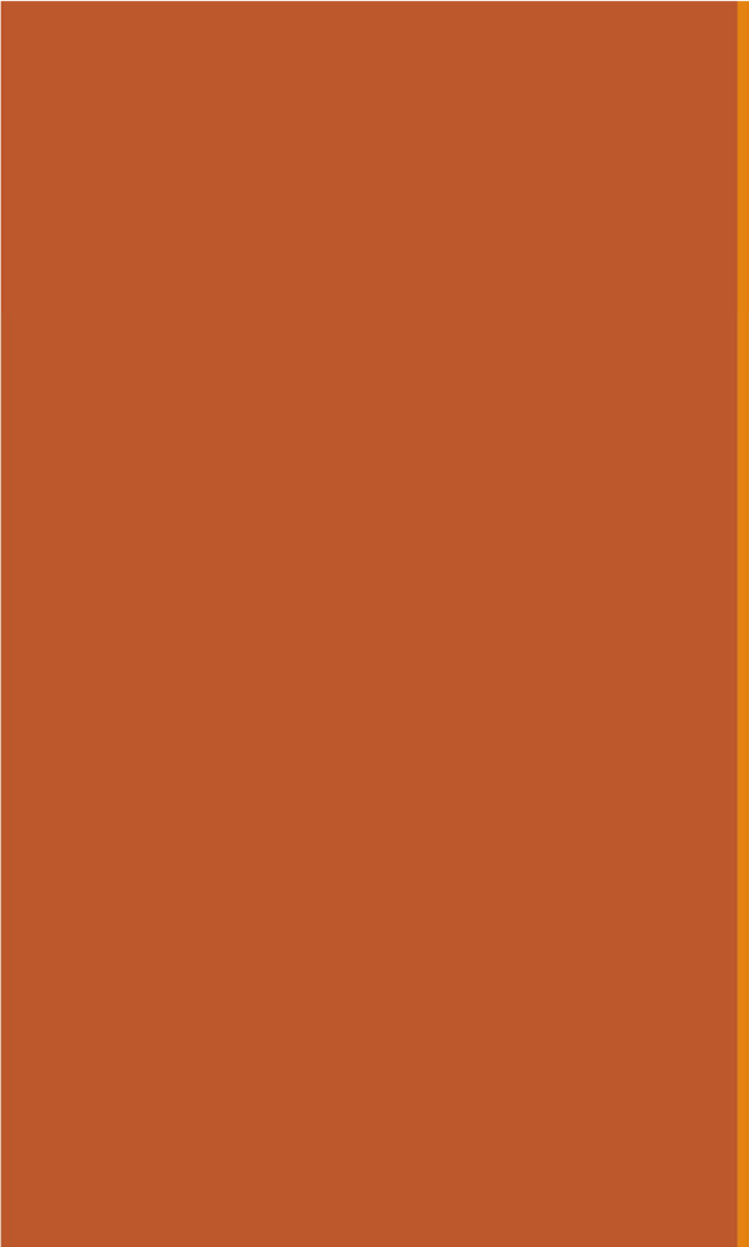
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14: Conditional Expectation

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[Lecture Discussion on Ed](#)



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Discrete conditional distributions

Discrete conditional distributions

Recall the definition of the conditional probability of event E given event F :

$$P(E|F) = \frac{P(EF)}{P(F)}$$

For discrete random variables X and Y , the **conditional PMF** of X given Y is

$$P(X = x | Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$



Different notation,
same idea:

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

Discrete probabilities of CS109

must be that $\sum_y \sum_t P(Y=y, T=t) = 1$

Each student responds with:

Year Y

- 1: Freshmen and Sophomores
- 2: Juniors and Seniors
- 3: Graduate Students and SCPD

Mood T :

- -1: 😞
- 0: 😐
- 1: 😊

Joint PMF

	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.06	.01	.01
$T = 0$	.29	.14	.09
$T = 1$	.30	.08	.02

$P(Y = 3, T = 1)$

this column is concerned
with the world where
everyone is a frosh
or sophomore

this row is concerned
with just those feeling
meh! (not good (not bad))

Joint PMFs sum to 1.

Discrete probabilities of CS109

The below are conditional probability tables for conditional PMFs

(A) $P(Y = y|T = t)$ and (B) $P(T = t|Y = y)$.

- Which is which?
- What's the missing probability?

	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.09	.04	.08
$T = 0$	.45	.61	.75
$T = 1$	.46	.35	.17

each column sums to 1... what does that mean?

	Joint PMF		
	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.06	.01	.01
$T = 0$	.29	.14	.09
$T = 1$	.30	.08	.02

normalize

	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.75	.125	?
$T = 0$	.56	.27	.17
$T = 1$	.75	.2	.05

these two rows sum to 1 why???



Discrete probabilities of CS109

The below are conditional probability tables for conditional PMFs

(A) $P(Y = y|T = t)$ and (B) $P(T = t|Y = y)$.

- Which is which?
- What's the missing probability?

	Joint PMF		
	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.06	.01	.01
$T = 0$	.29	.14	.09
$T = 1$	.30	.08	.02

(B) $P(T = t|Y = y)$

	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.09	.04	.08
$T = 0$	.45	.61	.75
$T = 1$	.46	.35	.17

(A) $P(Y = y|T = t)$

	$Y = 1$	$Y = 2$	$Y = 3$
$T = -1$	.75	.125	.125
$T = 0$	.56	.27	.17
$T = 1$	.75	.2	.05

1-.75-.125

normalizes world so that everyone is happy

.30 / (.06 + .29 + .30)

normalizes world to consist of juniors and seniors

Conditional PMFs also sum to 1 conditioned on different events!

Quick check

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

Number or function?

True or false?

1. $P(X = 2|Y = 5)$

number

5. $\sum_x P(X = x|Y = 5) = 1$ true

2. $P(X = x|Y = 5) = \frac{P(X=x, Y=5)}{P(Y=5)}$

1-D function of x

6. $\sum_y P(X = 2|Y = y) = 1$ false

3. $P(X = 2|Y = y) = \frac{P(X=2, Y=y)}{P(Y=y)}$

1-D function of y

7. $\sum_x \sum_y P(X = x|Y = y) = 1$

false

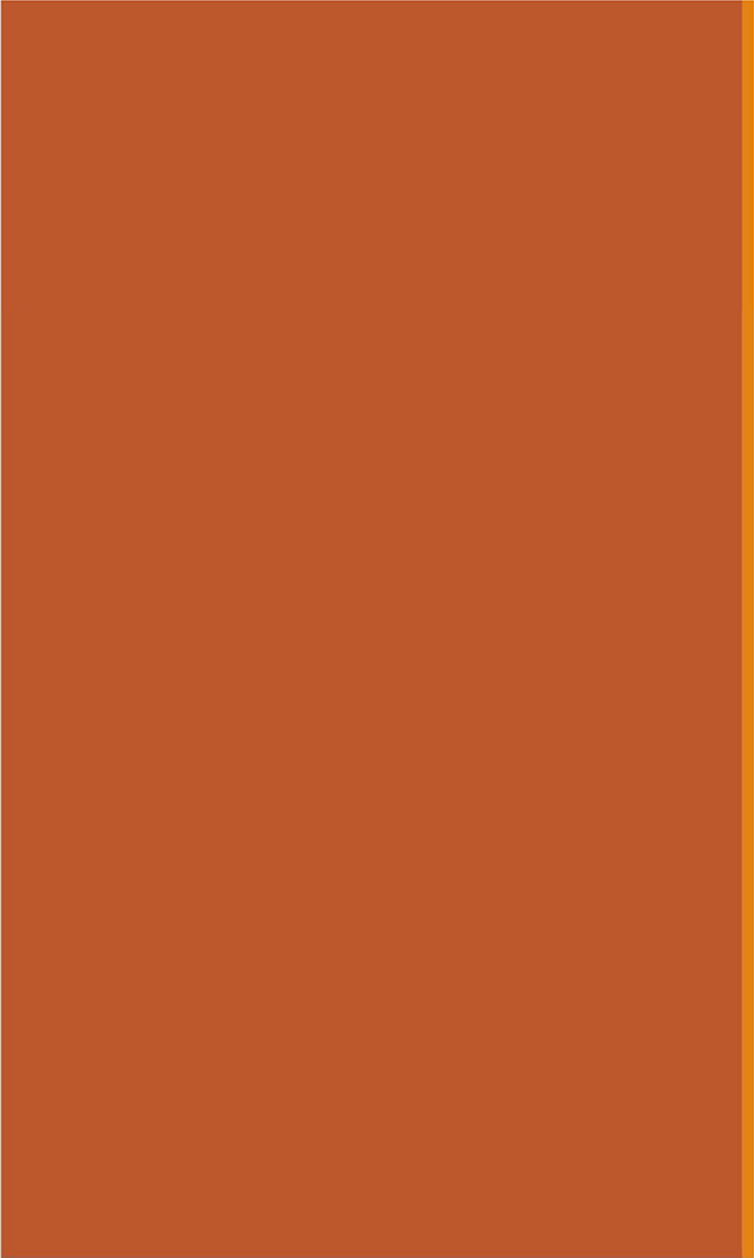
4. $P(X = x|Y = y) = \frac{P(X=x, Y=y)}{P(Y=y)}$

2-D function

of x, y

8. $\sum_x \left(\sum_y \underbrace{P(X = x|Y = y)P(Y = y)}_{\text{this is } P(X=x, Y=y)} \right) = 1$

true

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Conditional Expectation

Conditional expectation

discrete, though continuous is defined similarly

Recall the conditional PMF of X given $Y = y$:

$$p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

The **conditional expectation** of X given $Y = y$ is

$$E[X|Y = y] = \sum_x x \underbrace{P(X = x|Y = y)}_{p_{X|Y}(x|y)} = \sum_x x p_{X|Y}(x|y)$$

$Y=y$
is the
new
sample
space

It's been so long, our dice friends

$$E[X|Y = y] = \sum_x x p_{X|Y}(x|y)$$

- Roll two 6-sided dice.
- Let roll 1 be D_1 , roll 2 be D_2 .
- Let $S = \text{value of } D_1 + D_2$.



1. What is $E[S|D_2 = 6]$? *$S|D_2=6$ is a random variable!*

$$E[S|D_2 = 6] = \sum_{x=7}^{12} x P(S = x|D_2 = 6)$$
$$= \left(\frac{1}{6}\right) (7 + 8 + 9 + 10 + 11 + 12)$$
$$= \frac{57}{6} = 9.5$$

Intuitively: ✓ $6 + E[D_1] = 6 + 3.5 = 9.5$

P. 14
→ Let's prove this!
in a few slides
Stanford University 11

Properties of conditional expectation

1. LOTUS:

$$E[g(X)|Y = y] = \sum_x g(x)p_{X|Y}(x|y)$$

*Y=y is the
new sample
space*

2. Linearity of conditional expectation:

$$E\left[\sum_{i=1}^n X_i \mid Y = y\right] = \sum_{i=1}^n E[X_i|Y = y]$$

3. Law of total expectation (in, like, three slides)

CLIFF HANGER !!!

It's been so long, our dice friends

$$E[X|Y = y] = \sum_x x p_{X|Y}(x|y)$$

- Roll two 6-sided dice.
- Let roll 1 be D_1 , roll 2 be D_2 .
- Let $S = \text{value of } D_1 + D_2$.



1. What is $E[S|D_2 = 6]$?

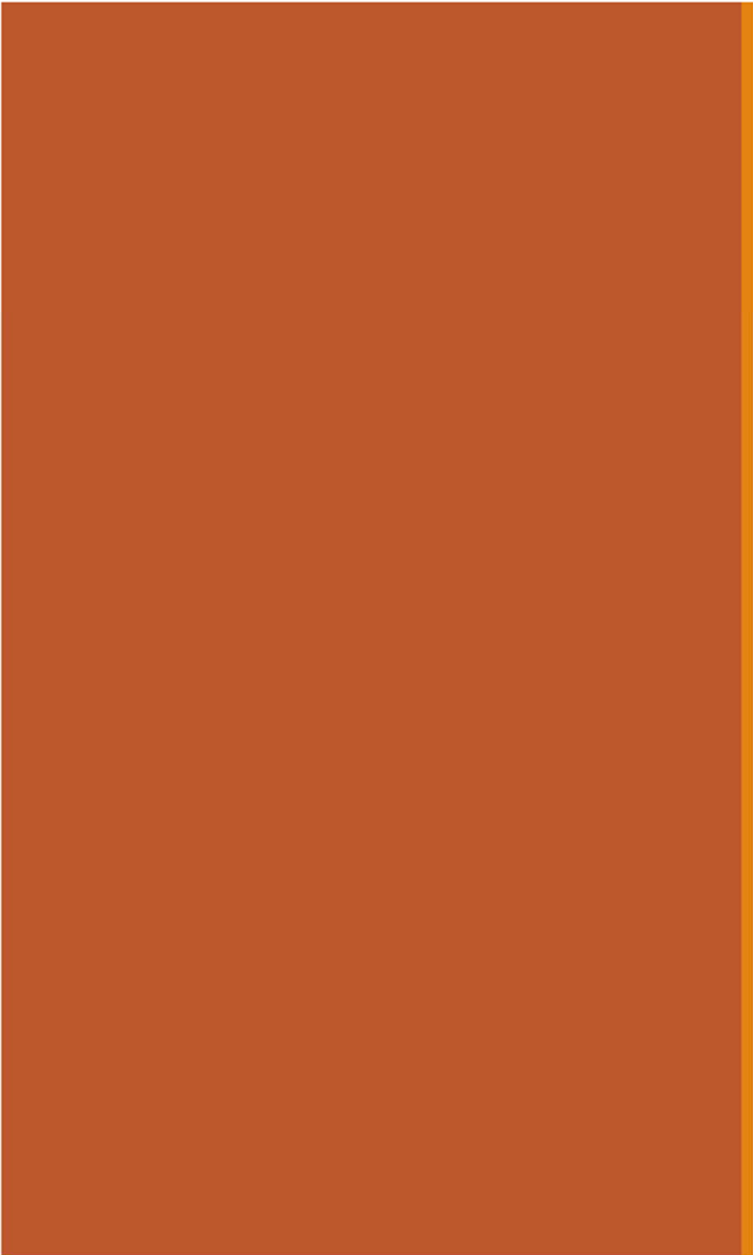
$$\frac{57}{6} = 9.5$$

2. What is $E[S|D_2]$?

- A. A function of S
 - B. A function of D_2**
 - C. A number
3. Give an expression for $E[S|D_2]$.

$$\begin{aligned} E[S|D_2 = d_2] &= E[D_1 + d_2|D_2 = d_2] \\ &= \sum_{d_1} (d_1 + d_2) P(D_1 = d_1|D_2 = d_2) \\ &= \sum_{d_1} d_1 P(D_1 = d_1) + d_2 \sum_{d_1} P(D_1 = d_1) \quad \begin{array}{l} (D_1 = d_1, D_2 = d_2 \\ \text{independent} \\ \text{events}) \end{array} \\ &= E[D_1] + d_2 = 3.5 + d_2 \end{aligned}$$

Handwritten notes:
- Red arrows point from d_1 and d_2 in the second line to the corresponding terms in the third line.
- The sum $\sum_{d_1} P(D_1 = d_1)$ is underlined in red with the note "this sum is just 1".
- The final result is $E[S|D_2] = 3.5 + D_2$.



Law of Total Expectation

Properties of conditional expectation

1. LOTUS:

$$E[g(X)|Y = y] = \sum_x g(x)p_{X|Y}(x|y)$$

2. Linearity of conditional expectation:

$$E\left[\sum_{i=1}^n X_i \mid Y = y\right] = \sum_{i=1}^n E[X_i | Y = y]$$

3. Law of total expectation:

$$E[X] = E[E[X|Y]] \quad \text{what?!}$$

this is a function on y

Proof of Law of Total Expectation

$$E[X] = E[E[X|Y]]$$

$$\begin{aligned}
 E[E[X|Y]] &= E[g(Y)] = \sum_y P(Y = y) E[X|Y = y] && (\text{LOTUS, } g(Y) = E[X|Y]) \\
 &= \sum_y P(Y = y) \sum_x x P(X = x|Y = y) && \begin{array}{l} \text{inner sum is just} \\ \text{definition of conditional} \\ \text{expectation} \end{array} \\
 &= \sum_y \left(\sum_x x P(X = x|Y = y) P(Y = y) \right) = \sum_y \left(\sum_x x P(X = x, Y = y) \right) && \begin{array}{l} \text{(def of conditional} \\ \text{expectation)} \end{array} \\
 &= \sum_y \left(\sum_x x P(X = x, Y = y) \right) = \sum_x \left(\sum_y x P(X = x, Y = y) \right) && \begin{array}{l} \text{(chain rule)} \\ \text{this is a constant} \\ \text{with respect to } x \end{array} \\
 &= \sum_x \sum_y x P(X = x, Y = y) = \sum_x x \sum_y P(X = x, Y = y) && \begin{array}{l} \text{(switch order of} \\ \text{summations)} \end{array} \\
 &= \sum_x x P(X = x) && \begin{array}{l} \text{this inner sum defines} \\ \text{the marginal prob mass} \\ \text{function of } x!! \end{array} \\
 &= E[X] && \text{...what?} \quad \text{(marginalization)}
 \end{aligned}$$

Another way to compute $E[X]$

$$E[X] = E[E[X|Y]]$$

$$E[E[X|Y]] = \sum_y P(Y = y)E[X|Y = y] = E[X]$$

If we only have a conditional PMF of X on some discrete variable Y , we can compute $E[X]$ as follows:

An assumption here is that it's difficult to compute marginal PDF of X .

1. Compute expectation of X given some value of $Y = y$
2. Repeat step 1 for all values of Y
3. Compute a weighted sum (where weights are $P(Y = y)$) *of all conditional expectations of X , i.e., $E[X|Y=y] \cdot y$*

```
def recurse():  
    if random.random() < 0.5:  
        return 3  
    else: return 2 + recurse()
```

Useful for analyzing recursive code.

Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y = y]P(Y = y)$$

```
def recurse():  
    # equally likely values 1,2,3  
    x = np.random.choice([1,2,3])  
    if x == 1: return 3  
    elif x == 2: return 5 + recurse()  
    else: return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y = y]P(Y = y)$$

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def recurse():  
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    if x == 1: return 3  
    elif x == 2: return 5 + recurse()  
    else: return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

$$E[Y] = E[Y|X = 1]P(X = 1) + E[Y|X = 2]P(X = 2) + E[Y|X = 3]P(X = 3)$$

$$E[Y|X = 1] = 3$$

When $X = 1$, return 3.

Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y=y]P(Y=y) \quad \text{If } Y \text{ discrete}$$

```
def recurse():  
    # equally likely values 1,2,3  
    x = np.random.choice([1,2,3])  
    if x == 1: return 3  
    elif x == 2: return 5 + recurse()  
    else: return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

$$E[Y] = E[Y|X=1]P(X=1) + E[Y|X=2]P(X=2) + E[Y|X=3]P(X=3)$$

$$E[Y|X=1] = 3$$

What is $E[Y|X=2]$?

- A. $E[5] + Y$
- B. $E[5 + Y] = 5 + E[Y]$
- C. $5 + E[Y|X=2]$



Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y=y]P(Y=y) \quad \text{If } Y \text{ discrete}$$

```
def recurse():  
    # equally likely values 1,2,3  
    x = np.random.choice([1,2,3])  
    if x == 1: return 3  
    elif x == 2: return 5 + recurse()  
    else: return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

$$E[Y] = E[Y|X=1]P(X=1) + E[Y|X=2]P(X=2) + E[Y|X=3]P(X=3)$$

$E[Y|X=1] = 3$

When $X=2$, return 5 +
a future return value of `recurse()`.

What is $E[Y|X=2]$?

- A. $E[5] + Y$
- ☒ B. $E[5 + Y] = 5 + E[Y]$
- C. $5 + E[Y|X=2]$

Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y = y]P(Y = y) \quad \text{If } Y \text{ discrete}$$

```
def recurse():  
    # equally likely values 1,2,3  
    x = np.random.choice([1,2,3])  
    if x == 1: return 3  
    elif x == 2: return 5 + recurse()  
    else: return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

$$E[Y] = E[Y|X = 1]P(X = 1) + E[Y|X = 2]P(X = 2) + E[Y|X = 3]P(X = 3)$$

$$E[Y|X = 1] = 3$$

$$E[Y|X = 2] = E[5 + Y]$$

When $X = 3$, return
7 + a future return value
of `recurse()`.

$$E[Y|X = 3] = E[7 + Y]$$

Analyzing recursive code

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y=y]P(Y=y) \quad \text{If } Y \text{ discrete}$$

```
def recurse():  
    # equally likely values 1,2,3  
    x = np.random.choice([1,2,3])  
    if x == 1: return 3  
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    else: return 7 + recurse()
```

Let Y = return value of `recurse()`.
What is $E[Y]$?

$$E[Y] = E[Y|X=1]P(X=1) + E[Y|X=2]P(X=2) + E[Y|X=3]P(X=3)$$

$$\uparrow$$
$$E[Y|X=1] = 3$$

$$\uparrow$$
$$E[Y|X=2] = E[5 + Y]$$

$$\uparrow$$
$$E[Y|X=3] = E[7 + Y]$$

$$E[Y] = 3(1/3) + (5 + E[Y])(1/3) + (7 + E[Y])(1/3)$$

$$E[Y] = (1/3)(15 + 2E[Y]) = 5 + (2/3)E[Y]$$

$$E[Y] = 15$$

On your own: What is $\text{Var}(Y)$?

Independent RVs, defined another way

If X and Y are independent discrete random variables, then $\forall x, y$:

$$P(X = x | Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} \overset{\text{red arrow}}{=} \frac{P(X = x)P(Y = y)}{P(Y = y)} = P(X = x)$$

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x, y)}{p_Y(y)} = \frac{p_X(x)p_Y(y)}{p_Y(y)} = p_X(x)$$

Note for conditional expectation, independent X and Y implies

$$E[X | Y = y] = \sum_x x p_{X|Y}(x|y) = \sum_x x p_X(x) = E[X]$$

Random number of random variables

Suppose you have a website: **zerothworldproblems.com**. Let:

- X = # of people per day who visit your site. $X \sim \text{Bin}(100, 0.5)$
- Y_i = # of minutes spent per day by visitor i . $Y_i \sim \text{Poi}(8)$
- ✓ X and all Y_i are independent.

The time spent by all visitors per day is $W = \sum_{i=1}^X Y_i$. What is $E[W]$?

$$E[W] = E\left[\sum_{i=1}^X Y_i\right] = E\left[E\left[\sum_{i=1}^X Y_i \mid X\right]\right]$$

$$= E[XE[Y_i]]$$

$$= E[Y_i]E[X] \quad (\text{scalar } E[Y_i])$$

$$= 8 \cdot 50$$

Suppose $X = x$.

$$E\left[\sum_{i=1}^x Y_i \mid X = x\right] = \sum_{i=1}^x E[Y_i \mid X = x] \quad (\text{linearity})$$

$$= \sum_{i=1}^x E[Y_i] \quad (\text{independence})$$

$$= xE[Y_i]$$